ARASH VASHAGH

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Summary

Machine Learning Developer completing an MSc in computer science with a research focus on adversarial machine learning. Published researcher with hands-on projects and several certifications from Google, Microsoft, AWS, and CompTIA. Available for full-time work and open to relocation across Canada for on-site work.

Skills

Programming: Python, Java, Kotlin, JavaScript, SQL, C++
AI/ML: PyTorch, TensorFlow, Agentic Design, RAG/LLMOps, Scikit-learn, Reinforcement Learning, Stable-Baselines3, Gymnasium, NLP
Web, Backend & APIs: React, HTML, CSS, Streamlit, FastAPI, Flask, REST APIs, OpenAI API, JSON Processing
Cloud & Deployment: AWS, Microsoft Azure, Google Cloud, Streamlit Community Cloud, Docker
Tools & Libraries: NumPy, Pandas, Matplotlib, PIL, Jupyter, VS Code, IntelliJ IDEA, Postman, PgAdmin 4, GitHub, MLflow
Languages: Persian (Native), English (Professional)

Certifications

- **AI and Machine Learning:** AWS Certified AI Practitioner; Microsoft AI-900; DeepLearning.AI Agentic AI
- **Cybersecurity:** CompTIA Security+; Microsoft SC-900
- **Cloud Computing:** AWS Certified Cloud Practitioner; Microsoft AZ-900; Google Cloud Digital Leader
- **Data and Project Management:** CompTIA Data+; Microsoft DP-900; CompTIA Project Management Essentials; CompTIA DataSys+ (results pending)

Experience

Graduate Research Assistant / Team Lead — UNB — Fredericton, Canada **Sep 2024 – Present**

- Conducting MSc thesis research in Adversarial Machine Learning, focusing on attacks, detection, and trustworthy AI.
- Co-authored ConformalShift, a targeted event-reordering attack against adaptive ECG monitoring systems.
- Co-authored a survey on adversarial attacks and another survey on adversarial defense mechanisms.
- Proposed and implemented two adversarial detection mechanisms, currently under internal faculty review.
- Developing an adversarial attack on fusion models, with manuscript in progress.
- Coordinating graduate student work and managing the Trustworthy and Secure AI (TSAI) LinkedIn page for lab updates and educational content.

Graduate Teaching Assistant — UNB — Fredericton, Canada **Jan 2025 – Apr 2026**

- *Software Engineering* (Winter 2026) – Java
- *Data Communication and Network Modeling* (Fall 2025) – Python
- *Data Science for Big Data Analytics* (Winter 2025) – Python

Conference Reviewer — Remote **Jul 2024 – Oct 2024**

- Reviewed manuscripts and provided structured feedback at international conferences:
  - *ArtInHCI 2024* – Artificial Intelligence and Human-Computer Interaction (Kunming, China) [Certificate]
  - *BIBE 2024* – Biological Information and Biomedical Engineering (Hohhot, China) [Certificate]

Research Assistant — IUT — Isfahan, Iran **Feb 2021 – Feb 2024**

- Conducted research on machine learning and healthcare, resulting in two IEEE publications [1] and [2].
- Collaborated with faculty on implementation, paper writing, and presenting at conferences.

Teaching Assistant — IUT — Isfahan, Iran **Feb 2021 – Feb 2024**

- Courses: Artificial Intelligence, Embedded Systems, Data Structures, and Programming Labs in C++, Java, and Python.
- Gave tutorials and graded assignments.

Education

Master's Candidate — University of New Brunswick (UNB), Fredericton, Canada **Sep 2024 – Aug 2026**

- Awarded Graduate Scholarship and Research Grant.
- Major: Computer Science
- Research Focus: Adversarial Machine Learning

Bachelor of Science — Isfahan University of Technology (IUT), Isfahan, Iran **Sep 2019 – Feb 2024**

- Major: Computer Engineering
- Minor: Intelligent Systems

Selected Projects

ConformalShift: Event-Reordering Attacks on Adaptive ECG Monitoring **Jun 2026 – Aug 2026**

- Developed a targeted attack against adaptive ECG monitoring systems that learn from feedback provided by doctors.
- Demonstrated that changing only the order of authentic feedback can reduce the detection of clinically important abnormal heartbeats.
- Designed and evaluated the attack using ECG datasets and multiple machine learning classifiers.
- Technologies: Python, Scikit-learn, adaptive conformal prediction, ECG analysis, healthcare AI.
- Paper: arXiv | Source code: GitHub.

PromptShield: Adversarial Detection for Vision-Language Models **Jan 2026 – Aug 2026**

- Developed a machine learning framework to detect adversarial images in vision-language models.
- Evaluated detection performance across multiple datasets, model backbones, and adversarial attack settings.
- Designed experiments for robustness evaluation, attack transferability analysis, and detector performance comparison.
- Built reproducible ML experiments for model training, evaluation, metric tracking, and result analysis.
- Technologies: Python, PyTorch, OpenCLIP, NumPy, Scikit-learn, adversarial machine learning.
- Manuscript and code under review at NeurIPS 2026.
- Source code: GitHub.

PaperWise AI: Deployed Agentic Research Paper RAG / LLMOps App **Jun 2026**

- Built and deployed a public Streamlit app for research paper question answering.
- Implemented PDF ingestion, text extraction, chunking, TF-IDF retrieval, OpenAI API integration, and offline fallback.
- Applied agentic design through suggested questions, reviewer critique mode, structured paper summaries, multi-paper comparison, and PDF answer export.
- Displayed RAG pipeline metrics including retrieval scores, response latency, retrieved chunks, and citation usage.
- Technologies: Python, Streamlit, Scikit-learn, OpenAI API, pypdf, ReportLab, Streamlit Community Cloud.
- Live app: Streamlit | Source code: GitHub.

Voluntary Educator (AI Security)

AI Security and Adversarial Machine Learning (AML) Course — Online **Jun 2026 – Present**

- Creating a Persian technical course on AI Security and AML to support learners facing language barriers.
- Developed slides and scripts on machine learning foundations for AML.
- Published course videos through a structured YouTube playlist and shared course materials on GitHub.